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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Introduction to Database Systems (course)

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# Thank you for taking the Week 4 : Assignment 4.



# Week 4 : Assignment 4

**Your last recorded submission was on 2025-02-19, 20:23 Due date: 2025-02-19, 23:59 IST. IST**

1) Which of the following relational query languages have the same expressive power? ***1 point***

I) Relational Algebra
II) Tuple relational calculus restricted to safe expressions
III) Relational Algebra with Select, Project, Cross-Product, Union and Difference operators only

- ○ II and III only
- ○ I and II only
- ○ I and III only
- ● I, II and III

2) What is the relational algebra expression equivalent to the following tuple relational calculus expression, where r is a relation name? ***1 point***

{ t | r(t) ∧ ( t.A = 10 ∧ t.B = 20) }

- ○ $\sigma_{(A=10 \vee B=20)}(r)$
- ○ $\sigma_{(A=10)}(r) \cup \sigma_{(B=20)}(r)$
- ● $\sigma_{(A=10)}(r) \cap \sigma_{(B=20)}(r)$
- ○ $\sigma_{(A=10)}(r) - \sigma_{(B=20)}(r)$

3) Consider the following two statements and choose the correct option: ***1 point***

S1: Each record of the referencing relation can relate to at most one record of the referenced relation.

S2: Each record of the referenced relation can relate to at most one record of the referencing relation.

- ◯ S1 is true, S2 is true
- ◯ S1 is false, S2 is true
- ◯ S1 is false, S2 is false
- ◉ S1 is true, S2 is false

4) Given the following relational schema **_1 point_**

Employee( <u>EmpID</u>, EmpName, Salary, Age, Address, Department, HireDate)

Which of the following TRC query retrieves employee id where the employee's salary is greater than ₹ 60,000, the employee's age is less than 40, the employee's department is either "IT" or "Marketing", and the employee's hire date is after January 1st, 2020.

Choose the correct option:

- ◯ { t.EmpID | Employee(t) ∧ t.Salary > 60000 ∧ t.Age < 40 ∧ (t.Department = "IT" ∨ t.Department = "Marketing") ∧ t.HireDate > '01/01/2020'}
- ◯ { t.EmpID | Employee(t) ∧ t.Salary > 60000 ∨ t.Age < 40 ∨ (t.Department = "IT" ∧ t.Department = "Marketing") ∨ t.HireDate > '01/01/2020'}
- ◉ { t.EmpID | Employee(t) ∧ (t.Salary > 60000 ∧ t.Age < 40) ∨ (t.Department = "IT" ∧ t.Department = "Marketing") ∧ t.HireDate > '01/01/2020'}
- ◯ { t.EmpID | Employee(t) ∧ (t.Salary > 60000 ∨ t.Age < 40) ∧ (t.Department = "IT" ∨ t.Department = "Marketing") ∧ t.HireDate > '01/01/2020'}

5) There are two weak entities E1 and E2, where E1 is the owner entity for E2. Consider relational representation of E1 and E2 and choose correct option: **_1 point_**

- ◉ Mapping of E1 should be done before E2
- ◯ Mapping of E2 should be done before E1
- ◯ E1 and E2 can be mapped in any order
- ◯ Partial key of E1 includes the partial key of E2

6) Consider the relation **_1 point_**
employee( <u>name</u>, sex, supervisorName )
The attribute supervisorName gives the name of the supervisor of the employee under consideration. What does the following Tuple Relational Calculus query produce?
{ e.name | employee(e) ∧
(∀x) [ ¬employee(x) ∨ x.supervisorName ≠ e.name ∨ x.sex = "male" ]}

- ◯ Names of employees with a male supervisor
- ◯ Names of employees with no immediate male supervisees
- ◉ Names of employees with no immediate female supervisees
- ◯ Names of employees with a female supervisor

7) Given the following relational schemas **_1 point_**
Student (<u>rollNo</u>, name, age, sex, deptNo, advisor)
Department (<u>deptId</u>, dName, hod, phoneNo)

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Which of the following will be the TRC query to obtain the department names that do not have any girl students?

- ( ) {d.dname | department(d) ∧ ¬((∃(s)) student(s) ∧ s.sex ≠ ‘F’ ∧ s.deptNo = d.deptId)}
- (x) {d.dname | department(d) ∧ ((∀(s)) student(s) ∧ s.sex ≠ ‘F’ ∧ s.deptNo = d.deptId)}
- ( ) {d.dname | department(d) ∧ ¬((∃(s)) student(s) ∧ s.sex = ‘F’ ∧ s.deptNo = d.deptId)}
- ( ) None of these.

8) Consider the relations S1( A, B, C ) and S2( C, D, E ) with primary keys A and C, respectively. The relation S1 contains 2000 tuples and S2 contains 2500 tuples. The maximum size of the join S1 $\bowtie_{S1.C = S2.C}$ S2 is ***1 point***

- ( ) 2000
- ( ) 2500
- (x) 4500
- ( ) 5000

9) Consider the following relational schema where rollNo and courseId are foreign keys in Enrollment referring to rollNo in Student and courseId in Courses, respectively: ***1 point***

Student( <u>rollNo</u>, name, degree, year, sex, deptNo, advisor )
Courses( <u>courseId</u>, cname, credits, deptNo )
Enrollment( <u>rollNo, courseId, sem, year</u>, grade )

Consider the TRC query given below:

{s.rollNo, s.name | Student(s) ∧ (∃e1)( Enrollment(e1) ∧ (e1.rollNo = s.rollNo)) }

Which one of the following is the correct interpretation of the above query?

- ( ) Retrieve rollNo and name of students who have enrolled for exactly one course
- ( ) Retrieve rollNo and name of students who have enrolled for more than one course
- ( ) Retrieve rollNo and name of students who have enrolled for at least one course
- (x) Retrieve rollNo and name of students who have all enrolled for same course

10) Consider the statements given below: ***1 point***

P: Tuple Relational Calculus (TRC) is a non-procedural query language
Q: Tuple Relational Calculus (TRC) expression specifies what is to be retrieved rather than how to retrieve it

Choose the correct option:

- ( ) P is true and Q is false
- (x) Both P and Q are true and Q is the correct reason for P
- ( ) Both P and Q are true and Q is not the correct reason for P
- ( ) Both P and Q are false

You may submit any number of times before the due date. The final submission will be considered for grading.

**Submit Answers**

Assessment submitted.
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